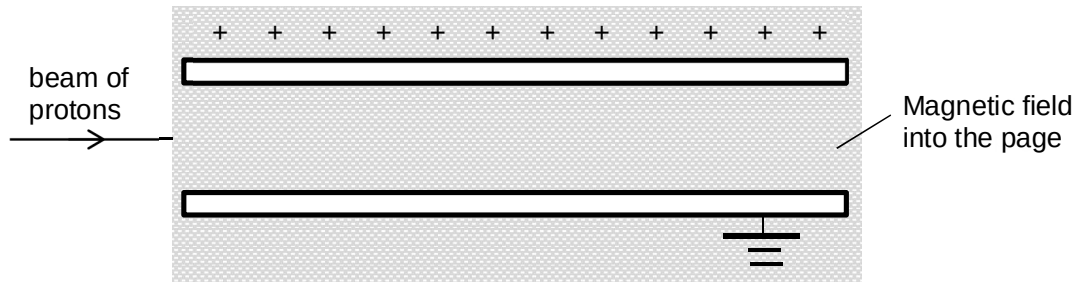


- 22** A velocity selector for protons has an electric field of strength $3.0 \times 10^5 \text{ V m}^{-1}$, produced by two horizontal plates. The upper plate is connected to a positive potential and the lower plate is earthed. A magnetic field of flux density $1.5 \times 10^{-2} \text{ T}$, directed into the plane of the paper, is at right-angle to the electric field as shown in the diagram.



Which of the following could be the possible speed of a proton and its observation?

	speed	observation
A	$1.8 \times 10^7 \text{ m s}^{-1}$	undeflected
B	$2.0 \times 10^7 \text{ m s}^{-1}$	deflected downwards
C	$2.2 \times 10^7 \text{ m s}^{-1}$	deflected downwards
D	$2.4 \times 10^7 \text{ m s}^{-1}$	deflected upwards